

DRY-RUN GUIDE | 15_vending_machine.cpp
Source: cpp\1st_belt\15_vending_machine.cpp

CORE BEHAVIOR

vendingMachine maps choices 1..5 to snack/exit strings and returns Invalid choice! for other values. main prints the returned string.

REPRESENTATIVE INPUT / CALL

Input: '3'.

STEP TRACE

switch matches case 3 and returns 'Spicy marshmallows'; main streams that return value.

EXPECTED OUTPUT / RETURN

'Spicy marshmallows'

ACTUAL SOURCE EXCERPT

Verbatim lines from the source; long lines wrap. Source files were not modified.

```
001 | /*
002 | Question:
003 | Write a program that takes the user's choice as input and prints the
004 | corresponding snack from the vending machine. If the choice is invalid,
005 | print "Invalid choice!".
006 |
007 | The vending machine has the following "delights":
008 |
009 | 1: Broccoli-flavoured ice cream
010 | 2: Chocolate-covered onions
011 | 3: Spicy marshmallows
012 | 4: Fizzy pickle juice
013 | 5: Exit (to save yourself from regret)
014 |
015 | Input Format:
016 | A single integer between 1 and 5 representing the user's choice.
017 |
018 | Output Format:
019 | The snack name corresponding to the selected choice, or "Invalid choice!"
020 | if the input is outside the range 1-5.
021 | */
022 |
023 | #include <iostream>
024 | #include <string>
025 | using namespace std;
026 |
027 | string vendingMachine(int choice) {
028 |     switch (choice) {
029 |         case 1:
030 |             return "Broccoli-flavoured ice cream";
031 |         case 2:
032 |             return "Chocolate-covered onions";
033 |         case 3:
034 |             return "Spicy marshmallows";
035 |         case 4:
036 |             return "Fizzy pickle juice";
037 |         case 5:
```

```
038 |         return "Exit";
039 |     default:
040 |         return "Invalid choice!";
041 |     }
042 | }
043 |
044 | int main() {
045 |     int choice;
046 |     cin >> choice;
047 |
048 |     cout << vendingMachine(choice);
049 |
050 |     return 0;
051 | }
052 |
```